DIGITAL DISCOURSE ON CLIMATE SECURITY: STRUCTURAL RESILIENCE AND EMOTIONAL PARADOXES DURING COP27

PRESENTATION

Presented by: **Any Das (922710) | Arnab Biswas (925349) | Summan Gul (925663)**
Università degli Studi di Milano-Bicocca
Social Media Analytics
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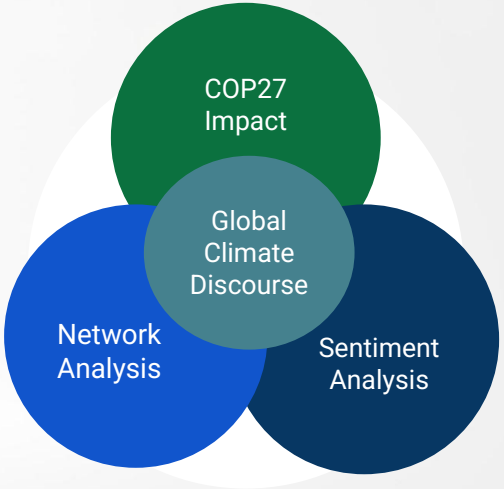
Project Overview & Objective

Core Focus: Analyzes "Climate Security" discourse on X (Twitter) during the COP27 Life Cycle (June 2022 – May 2023).

Aim: To understand how major summits shape structural and emotional dynamics.

Objectives:

1. **Structural Unity:** Determine if the global network is unified or fragmented.
2. **The Influence Paradox:** Uncover hidden dynamics where popularity ≠ influence.
3. **Event Impact:** Assess how COP27 triggers engagement and shapes public opinion.



Data Description

Source: "Climate Security on Twitter" dataset (Carneiro & Tucci).

Collection: Snowball Sampling technique used to capture evolving conversations.

Dataset Scope: ~636,379 unique tweets (Jan 2014 – May 2023)

Data Components:

1. Annotated Content Data: Tweet IDs, timestamps, hashtags, pre-calculated sentiment polarity (NRC), and country mentions.
2. Network Nodes Data: Hashtags with frequency and temporal activity.
3. Network Edges Data: Undirected weighted co-occurrence connections.

Category	Key Hashtags (Seed Queries)
Core Concepts	#ClimateSecurity, #PlanetarySecurity
Conflict & Risk	#WaterConflict, #ClimateFragility
Justice & Rights	#ClimateJustice, #ClimateMigration
Diplomacy	#SustainingPeace, #ClimateDiplomacy
Multilingual	#SecuriteClimatique (FR), #SeguridadClimatica (ES)

Data Pre-processing

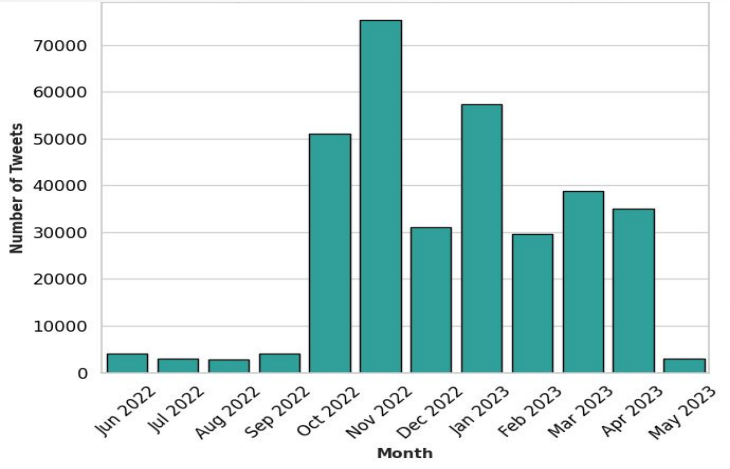
Time & Filtering: June 1, 2022 – May 31, 2023 (focus on COP27)

Handling Missing Data:

Sentiment: Filled missing values using hashtag-specific mean imputation.

Tags: Labeled missing hashtags as “NoHashtag” and countries as “Unknown”

Timestamps: Converted from Gephi format to standard datetime objects.



*Monthly tweet distribution showing the massive surge during **COP27***

After Preprocessing:

Annotated Tweets: 335,112 (from 636,379)

Nodes: 35,351 (Unique Hashtags)

Edges: 353,192 (Co-occurrence Links)

Methodology | Social Network Analysis

Framework: Constructed an undirected weighted graph using Python (NetworkX v2.8).

Network Hierarchy (K-Core): Revealed the layered "Onion Model" and resilient Deep Core.

Thematic Clustering (Louvain): Detected distinct conversation groups and key communities.

Influence Dynamics (Centrality): Measured Degree vs. Betweenness to distinguish "Bridging Hashtags" from popular ones.

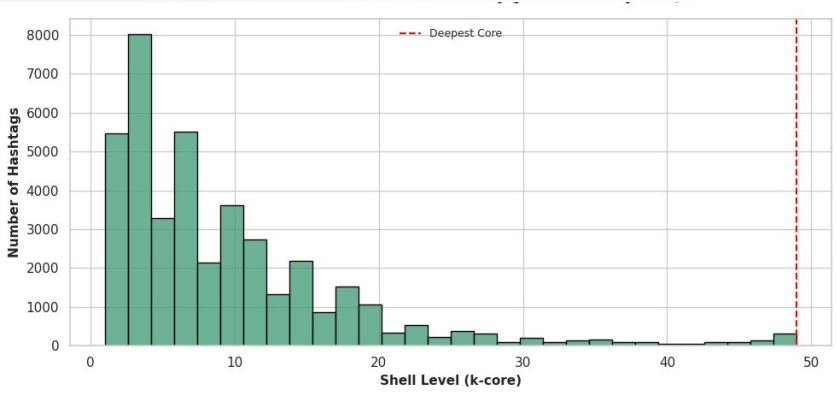
Polarization Test (Assortativity): Tested for structural democracy (openness) vs. content echo chambers.

Network Stability (Density): Tracked Weighted Density to quantify the "Dilution Effect" during COP27.

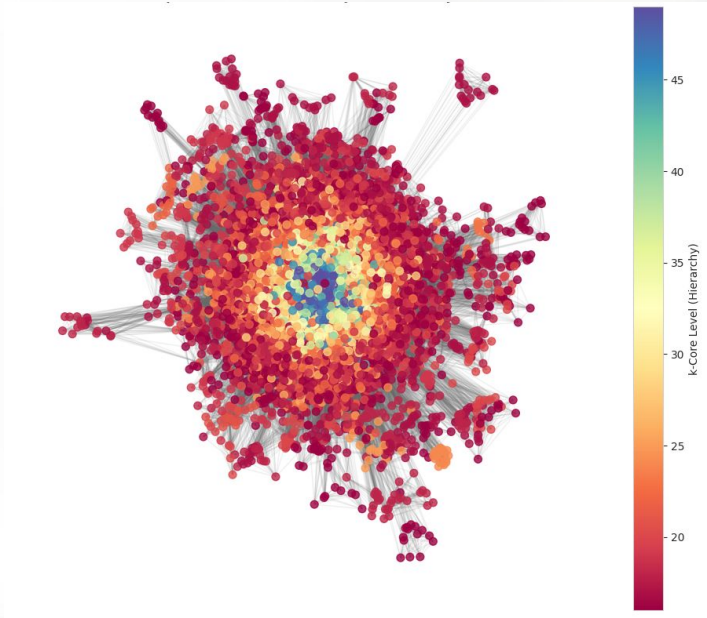
Network Topology: The “Onion Model”

The Periphery (k < 5): Massive outer layer of temporary, "one-off" participants.

The Deep Core (Max k = 49): A resilient nucleus of ~250 hashtags (e.g., #COP27, #ClimateCrisis) acting as the network's backbone.



Shell Distribution (Onion Model)

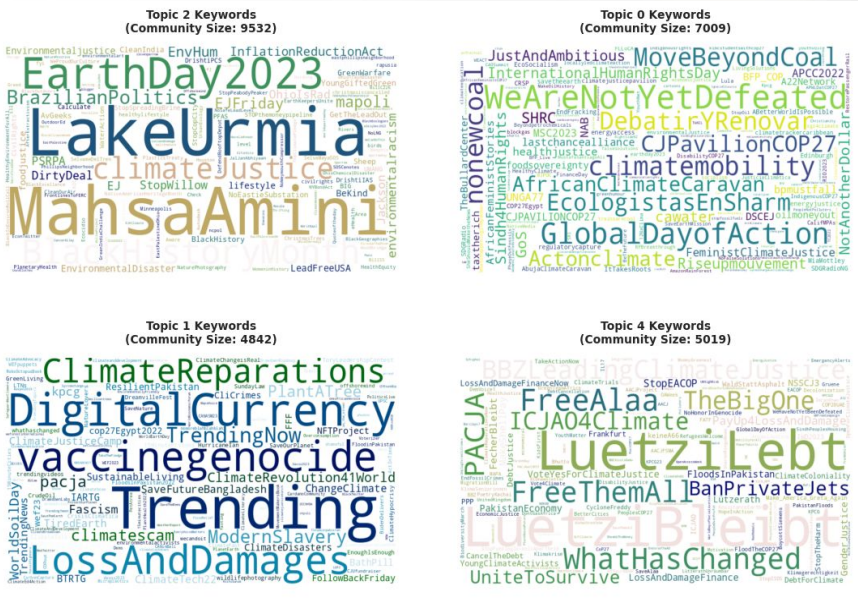


Deep Core Structure (k > 15)

Community Detection

Louvain algorithm shows high modularity (Q = 0.54) with 120 distinct communities, confirming a segmented discourse.

Topic	Thematic Focus	Key Hashtags
Topic 2	Human Rights & Justice (Intersectional Justice)	#MahsaAmini, #ClimateJustice
Topic 0	Global Activism (COP27) (Institutional Backbone)	#NoNewCoal, #GlobalDayofAction
Topic 1	Finance & Skepticism (The Friction Point)	#LossAndDamage ↔ #ClimateScam
Topic 4	Protests & Resistance (Grassroots Resistance)	#LuetziBleibt, #StopEACOP



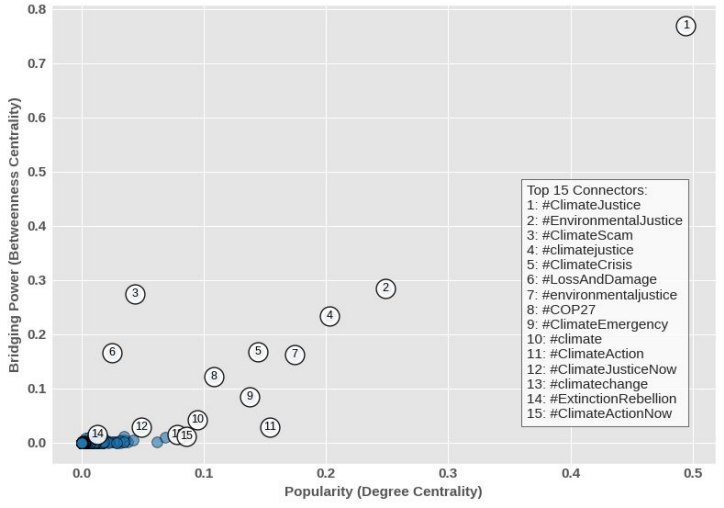
Top 4 Thematic Communities (Word Clouds)

The Connector Paradox

The Super-Connector: #ClimateJustice is the strongest link, proving "Justice" is the main foundation.

The Hidden Bridge (Anomaly): #ClimateScam has low popularity but acts as a powerful bridge for skeptics.

The Weak Bridge: #ClimateAction has high visibility but fails to connect communities (Echo Chamber).

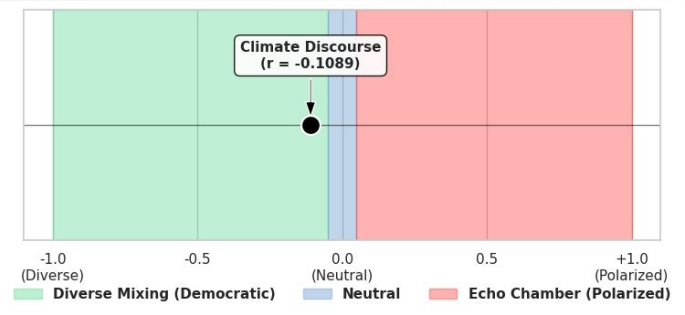


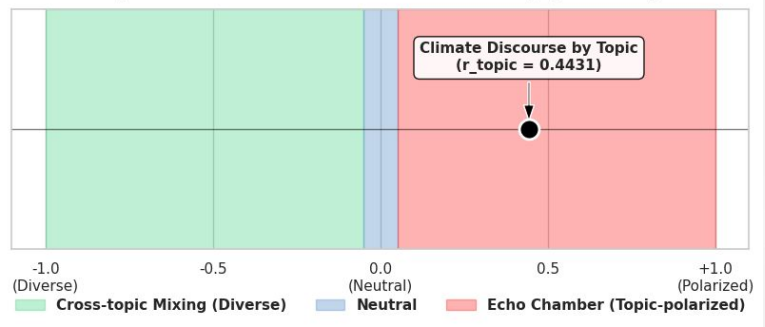
Popularity vs. Bridging Power

Hashtag	Degree (Popularity)	Betweenness (Influence)
#ClimateJustice	**0.4942**	**0.7699**
#ClimateScam	**0.0436**	**0.2741**
#LossAndDamage	0.0250	0.1657
#ClimateAction	0.1542	0.0293

The Structural–Thematic Paradox

The Structure (Network Topology)	The Content (Discourse Topics)
Metric: Structural Assortativity	**Metric:** Topic Assortativity
Score: -0.11 (Negative)	**Score:** +0.44 (Positive)
Pattern: "Democratic Core"	**Pattern:** "Content Bubbles"
Insight: Hubs connect everyone.	**Insight:** Topics stay isolated.





Comparative Analysis of Mixing Patterns

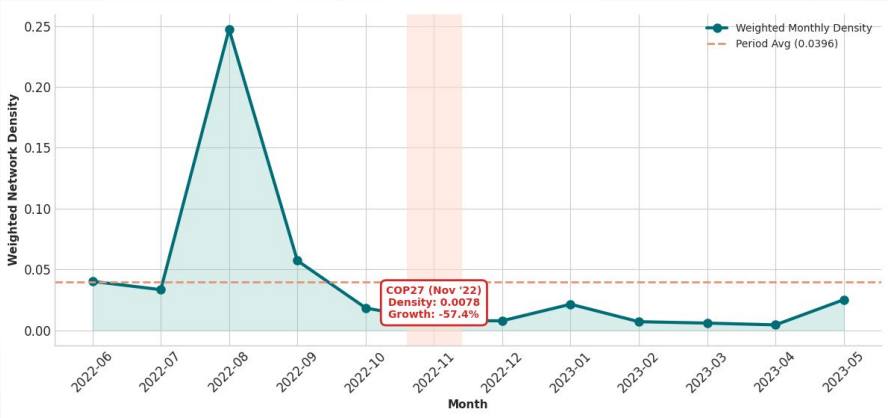
The Dilution Effect (Network Stability)

Maximum Cohesion Phase (August 2022): Network peaked at 0.25, driven by a dedicated activist core.

Structural Collapse (November 2022): COP27 triggered a 57.4% drop to 0.0078; tweet volume increased but cohesion collapsed.

The "Dilution Effect": Temporary "tourist" users broadcasting messages diluted core ties, fragmenting the network.

Slow Recovery (May 2023): Cohesion gradually returned after temporary crowd departed.



Monthly Weighted Density

Methodology | Social Content Analysis

Basis: Analysis derived from NRC Emotion Lexicon sentiment scores.

Geographic Profiling (Welch's T-test): Classified tweets (North vs. South) to assess the statistical significance of regional emotional differences.

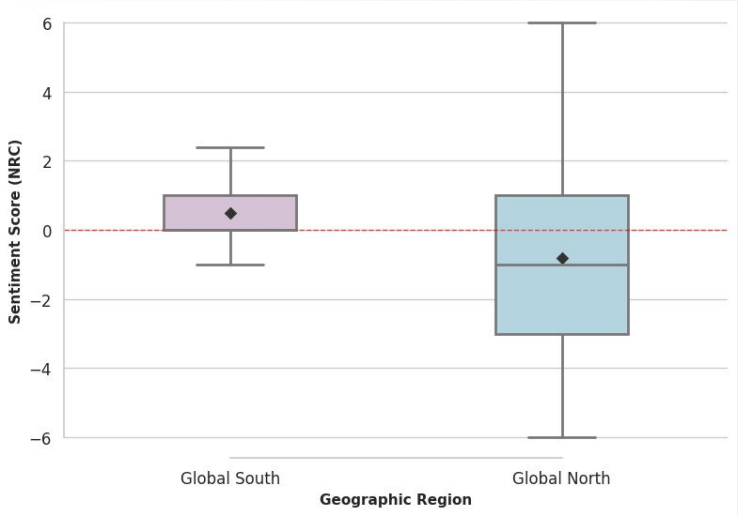
Temporal Dynamics: Applied a 7-day simple moving average to smooth noise across Pre, During, and Post-COP27 phases.

Volatility Analysis (SVI): Calculated daily standard deviation of sentiment scores to measure instability in public opinion..

Correlation Analysis (Spearman Rank): Quantified the relationship between discourse intensity and emotional tone.

The Resilience Paradox (Regional Analysis)

- **Emotional Divide:** Global North vs. Global South sentiment shows a sharp contrast.
- **Global North (n=1,533):**
  - Mean: -0.81 (Negative)
  - Wide variance → Polarized, skeptical discourse.
- **Global South (n=71,456):**
  - Mean: +0.49 (Positive)
  - Tightly clustered → Hopeful, solution-oriented.
- **Welch's T-test (T = -21.15)** confirms this disparity is statistically significant ($p < 0.001$), proving distinct regional emotional behaviors.



Polarization (North) vs. Consensus (South)

Temporal Dynamics (The COP27 Effect)

COP27 triggered a **+9.6% rise** in sentiment while volatility dropped, indicating a rare moment of **emotional consensus.**

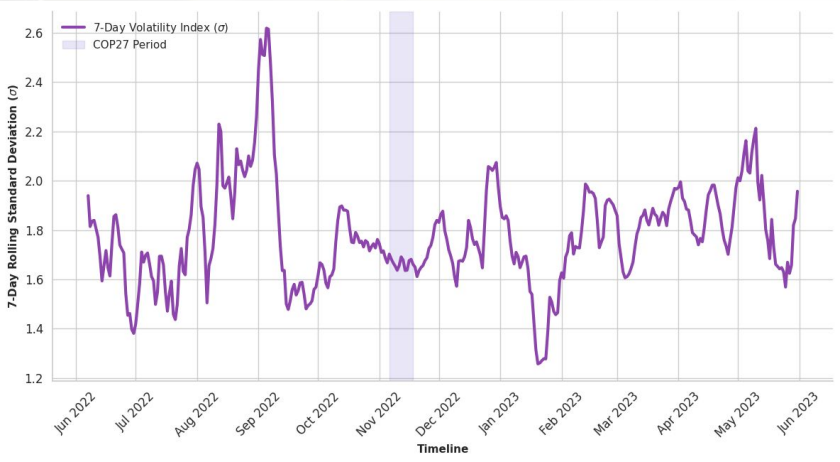
Post-Summit sentiment fell by **-14.1%**, dropping below the pre-summit baseline, proving that the positive momentum was temporary.

An absolute peak (1.96) occurred in late May 2023, driven by spring extreme weather events and Bonn preparatory meetings.

Phase	Count (n)	Mean Sentiment	Std. Deviation
Pre-Summit	74,882	0.7653	1.7950
During COP27	**47,360**	**0.8392**	**1.6940 (Low Volatility)**
Post-Summit	212,870	0.7210	1.7555

Sentiment Evolution: COP27 Surge & Decay

Discourse Stability (Volatility Analysis)



Sentiment Volatility Index (Crisis vs. Summit)

Peak Instability (Sept 2022): Volatility peaked at ~2.6, driven by overlapping crises (heatwaves, energy insecurity) rather than the summit itself.

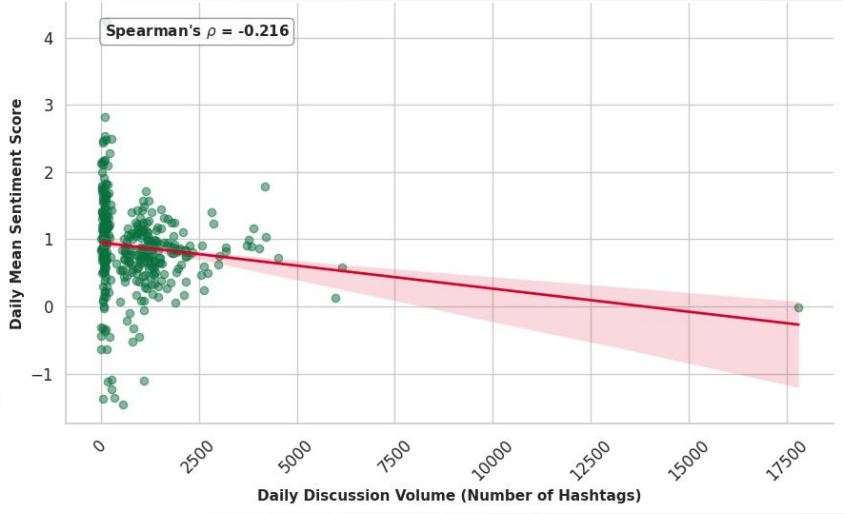
Structured Debate (During COP27): Volatility remained moderate (~1.6) despite high sentiment. This indicates focused debate rather than emotional chaos.

Maximum Stability (Jan 2023): Dropped to ~1.25, reflecting reduced activist mobilization after the summit.

Major geopolitical and climate crises generate sharper emotional instability than formal international summits.

The Volume-Sentiment Paradox

- Spearman Rank Analysis (ρ = **-0.216**) confirms a significant negative link: more discussion does not mean more positivity
- Mass engagement is driven by **"Negative Urgency"** (crisis response) rather than optimism, leading to lower average sentiment.
- Broader audiences dilute the positivity of the activist core, a phenomenon we define as the **"Dilution Effect"**.



Regression Plot: Volume vs. Sentiment

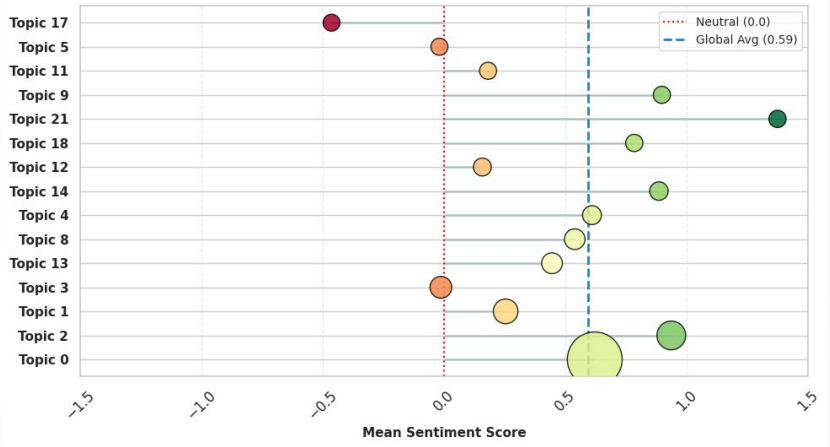
Community-Level Sentiment

Mapping Sentiment to Network Structure

Positive-Scale Communities (Topics 0, 2): Massive clusters like "Justice" (+0.93) and "COP27" (+0.62) maintain high positivity, effectively forming the backbone of the activist movement.

Positive-Niche Communities (Topic 21): Highest optimism (+1.37) exists in small, specialized clusters focused on solutions and solidarity.

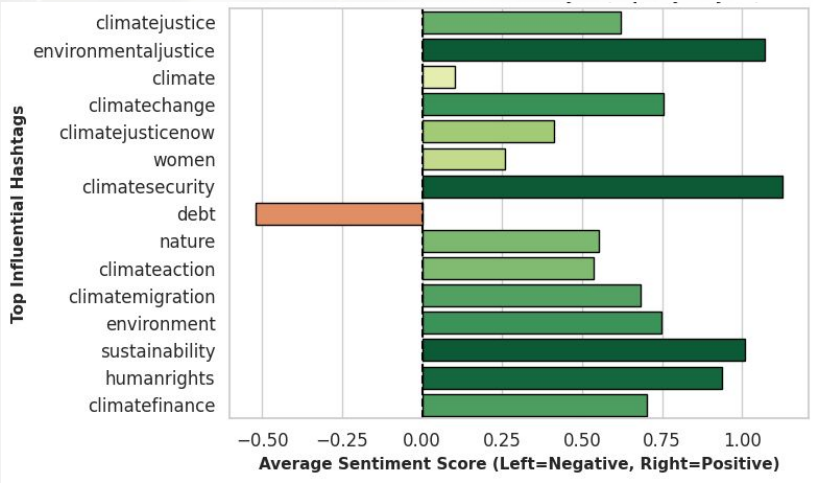
Critical Outliers (Topic 17): Negative sentiment (-0.46) is isolated in clusters dominated by grievance and crisis narratives.



Sentiment by Community Volume

Influencer-Level Sentiment

Sentiment of Structural Bridges



Influential Hashtags Ranked by Sentiment

- **#ClimateSecurity (+1.12)** and **#Sustainability (+1.00)**: Strongest positive bridges, framing security through solutions rather than fear.
- **#EnvironmentalJustice (+1.07)**: A pillar of trust and solidarity, connecting diverse discourse strands.
- **#Debt (-0.52)**: Primary negative outlier, channeling deep frustration regarding economic inequality and climate finance.

Solution-building flows through positive connectors, while financial tensions flow through critical ones.

Conclusion

The Anchor: #ClimateJustice serves as the universal bridge, connecting diverse communities from grassroots to policymakers.

Four Key Paradoxes:

- **Dilution Effect:** Big summits (COP27) increase visibility but weaken community cohesion, creating "Diluted Stability."
- **Resilience Paradox:** Vulnerable Global South leads constructive discourse, while the North remains skeptical.
- **Influence Paradox:** Low-volume tags (e.g., #ClimateScam) act as powerful hidden bridges, proving popularity ≠ influence.
- **Structural-Thematic Paradox:** Network structure is democratic, yet content remains locked in Topic Bubbles (Echo Chambers).

Friction Points: #Debt acts as a critical barrier, shifting conversations from hope to grievance.

THANK YOU

ANY QUESTIONS?